

Post-Quantum Secure Chameleon Hash Functions for Redactable Blockchains: A Survey

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Blockchain systems derive their security from immutability, yet legal frameworks such as the GDPR grant a right to erasure that an append-only ledger cannot structurally honour. The dominant cryptographic response is the *redactable blockchain*, built on a *chameleon hash function*, a trapdoor hash whose holder can compute collisions on demand, replacing on-chain content while leaving the surrounding chain intact. Almost all such constructions rest on assumptions which Shor’s algorithm breaks, and quantum resistance cannot be retrofitted to commitments already written to a chain.

This article surveys the intersection of post-quantum cryptography and redactable blockchain. Where existing surveys organise the field by redaction mechanism and trapdoor management, we take the post-quantum frontier as our subject and measure it against the classical state of the art. Across thirty-one papers, of which eight are post-quantum, the literature reaches parity on collision resistance and indistinguishability, holds shallow footholds on multi-authority operation, dynamic membership, and revocability, and has produced nothing with strong accountability, forward and backward security, policy privacy, or time-bound redaction; no construction is at once rigorously proved and empirically validated. We identify two pieces of classical machinery with no known lattice analogue, one of them accounting for several absent properties at once, and localise the epoch-transition problem to the layer at which the classical literature itself solves it. We close with a research agenda ordered by dependency, beginning with the enabling work the remaining gaps rest on.

CCS Concepts: • **Security and privacy** → **Cryptography; Symmetric cryptography and hash functions**; *Public key (asymmetric) techniques; Mathematical foundations of cryptography; Key management; Distributed systems security*; Access control.

Additional Key Words and Phrases: redactable blockchain, chameleon hash, post-quantum cryptography, quantum-resistant cryptography, lattice-based cryptography, Chameleon Hash with Ephemeral Trapdoors, Policy-Based Chameleon Hash, key exposure freeness, collision resistance, forward and backward security, accountability, revocability, policy privacy, GDPR, right to erasure, systematic literature review

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1 Introduction and Motivation

Blockchain security rests in large part on immutability: altering an appended record requires breaking the hash linkage binding every block to its predecessor. In a growing number of deployments this has become a liability. The European Union’s General Data Protection Regulation (GDPR) grants a right to erasure that an append-only ledger cannot

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structurally honour, and operational need presses in the same direction, as healthcare systems must correct erroneous records, financial institutions must rectify mistaken entries, and Internet of Things (IoT) deployments must discard sensor data that has outlived its value. Each requires modifying on-chain content selectively and verifiably, without disturbing the surrounding ledger.

The dominant response is the *redactable blockchain*, introduced in its modern form by Ateniese et al. [7]. Its mechanism is the *chameleon hash function* (CHF): a trapdoor hash whose holder can compute a collision on demand, replacing block or transaction content while leaving the hash value, and hence the rest of the chain, unchanged. Subsequent work has added attribute-based access control, threshold trapdoor management, accountability, and revocation.

Almost all of these constructions rest on discrete logarithm, RSA, or pairing assumptions, each efficiently solvable by Shor’s algorithm [42]. Ledger records are long-lived, and quantum resistance cannot be retrofitted to commitments already written to a chain, so a redactable blockchain intended for long-term deployment must be post-quantum from the outset. Lattice-based cryptography offers candidate replacements: the Short Integer Solution (SIS) and Learning With Errors (LWE) problems are believed to resist quantum attack, and the trapdoor machinery of Gentry, Peikert and Vaikuntanathan (GPV) [21] and Micciancio and Peikert [36] supplies the tools for building trapdoor hashes from them. The intersection of the two lines of work nonetheless remains largely unexplored.

The survey literature reflects this. Redactable blockchain has been reviewed several times, each along an axis orthogonal to cryptographic foundation: the tension between immutability and erasure law [40], redaction mechanism and evaluation criteria [60], permission model [1], hashing scheme and ledger structure [20], or research topic across the redaction lifecycle [32]. Quantum resistance is a dimension of comparison in none of them. Table 1 summarises this coverage. Timing compounds the gap, since roughly half of the post-quantum corpus postdates the search windows of all but the most recent prior review.

This article takes that intersection as its subject, and contributes:

- a structured comparison of the CHF and redactable blockchain literature (Tables 4–7) treating quantum resistance as a primary axis, and a tiered account of post-quantum standing by property (Table 8);
- a separation of incidental absences from structural ones, and a decomposition of the epoch-transition problem isolating the layer at which the classical mechanisms actually operate, accounting for several absent properties at once;
- a research agenda ordered by dependency, identifying for each gap the classical mechanism to be re-derived and the base construction best suited to it.

Table 1. Existing survey and review articles on redactable blockchain, and their treatment of post-quantum redaction. × = not addressed; ~ = post-quantum schemes cited but not used as a dimension of comparison.

Ref	Year	Scope and Contribution	PQ
[40]	2019	Immutability against the GDPR right to erasure; early proposals.	×
[60]	2021	Redaction mechanisms for mutable blockchains, and evaluation criteria.	×
[47]	2021	Eight revocation mechanisms, of which chameleon-hash redaction is one.	×
[59]	2023	Redactable blockchain literature, limitations, and open problems.	×
[1]	2023	Chameleon-hash and non-chameleon mechanisms, by permission model.	×
[20]	2023	CHF and CHET algorithms; classified by hashing scheme and block structure.	×
[32]	2026	Systematic review of 40 papers; ten research topics, twenty-one techniques.	~
This paper	2026	Post-quantum literature evaluated against the classical state of the art across twenty-six criteria, separating incidental absences from structural ones.	

2 Background & Preliminaries

This section establishes the cryptographic foundations on which the remainder of the survey builds. We assume familiarity with standard blockchain architecture and focus on the cryptographic primitives and hardness assumptions that are less likely to be known to a general blockchain audience.

2.1 Chameleon Hash Functions

In the redactable blockchain setting, *redaction* refers to the replacement of the content of a block or transaction with new content, such that the resulting hash value is identical to the original and the integrity of the surrounding chain is entirely preserved. The hash functions used for this purpose are *chameleon* hash functions. These are hash functions with a *trapdoor* key which, when known, allow us to compute a *collision*.

Definition 2.1. A collision for a hash function Hash is a pair of distinct inputs $(m, r) \neq (m', r')$ such that $\text{Hash}(m; r) = \text{Hash}(m'; r')$.

Definition 2.2 (Uniqueness). Hash is unique if for every message m and hash value h there is at most one r satisfying $\text{Hash}(m; r) = h$.

In a standard cryptographic hash function, collisions exist but are computationally infeasible to find. A Chameleon Hash Function (CHF) retains this infeasibility for any party lacking the trapdoor, while providing an authorised holder of the trapdoor key sk with an efficient algorithm for computing collisions on demand.

Definition 2.3. A Chameleon Hash Function (CHF) is a trapdoor hash scheme enabling this operation. Formally, a CHF consists of four algorithms:

- $\text{Gen}(1^\lambda) \rightarrow (pk, sk)$: Given a security parameter λ , outputs a public key pk and a secret trapdoor key sk .
- $\text{Hash}(pk, m; r) \rightarrow h$: Given a public key and message m with randomness r , outputs a hash value h .
- $\text{Ver}(pk, m, h, r) \rightarrow \{0, 1\}$: Verifies that (m, r) is a valid opening of h under pk .
- $\text{Adapt}(sk, m, m', h, r) \rightarrow r'$: Given the trapdoor, finds new randomness r' such that $\text{Hash}(pk, m'; r') = h$. This is the collision-finding operation.

We also require correctness: $\text{Ver}(pk, m', h, r') = 1$ whenever $r' \leftarrow \text{Adapt}(sk, m, m', h, r)$.

Collision Resistance Notions. The literature has accumulated several distinct notions of collision resistance for CHFs, which are not equivalent and have different implications for security in deployed systems. Intuitively, collision resistance captures the requirement that only an authorised trapdoor holder should be able to compute collisions. Following the taxonomy of Derler, Samelin and Slamanig [16], four notions are relevant to this survey:

- **Weak collision resistance (W-CR):** An adversary given only the public key, with no access to any collision, cannot find one.
- **Enhanced collision resistance (E-CR):** An adversary cannot find a collision for a hash on which it has seen no collision, even after obtaining collisions on other hashes. It gives no guarantee for a hash whose collision has already been observed. This is the notion introduced and relied upon by Ateniese et al. [7].
- **Standard collision resistance (S-CR):** An adversary cannot find a collision for a fresh message even after obtaining collisions on other messages, including collisions on the target hash itself.
- **Full collision resistance (F-CR):** An adversary cannot find any new collision even after adaptively querying a collision-finding oracle. This is the strongest notion and the correct security target for redactable blockchain

schemes in which collisions are publicly observable on-chain. Derler et al. [16] prove that F-CR strictly implies both E-CR and S-CR, and conversely that E-CR and S-CR together imply F-CR; since E-CR and S-CR are themselves incomparable, neither alone suffices.

Key Exposure Freeness. A CHF is *key exposure free* (KEF) if computing a collision (i.e., running Adapt) reveals no information about the trapdoor sk to any observer. In a live blockchain, collisions are broadcast publicly when a redaction occurs; without KEF, an adversary accumulating sufficient collisions could eventually recover the trapdoor and perform arbitrary unauthorised edits.

2.2 Chameleon Hash with Ephemeral Trapdoors

A *Chameleon Hash with Ephemeral Trapdoors* (CHET), introduced by Camenisch et al. [8], extends the standard CHF by requiring *two* secrets to compute a collision: the long-term trapdoor sk and a per-hash ephemeral trapdoor etd generated fresh by the hashing party at hash-computation time.

Only two of the four CHF algorithms change. $\text{Hash}(pk, m; r) \rightarrow (h, etd)$ additionally outputs the per-hash ephemeral trapdoor, which is retained by the party that computed the hash, and $\text{Adapt}(sk, etd, m, m', h, r) \rightarrow r'$ takes etd alongside the long-term trapdoor; collision-finding requires *both* sk and etd , neither alone being sufficient. Gen, Ver, and correctness are as above.

The key consequence is per-hash control over collision-finding: the hashing party receives a fresh etd from Hash and may subsequently share or withhold it at will. Even the long-term trapdoor holder cannot run Adapt on a hash without the corresponding etd , a property formalised as private collision resistance [8]. CHET is the foundational primitive underlying virtually all policy-based redactable blockchain schemes in the literature, where etd is stored as auxiliary ciphertext alongside the transaction so that only an authorised redactor can recover it.

Policy-Based Chameleon Hashes. A *Policy-Based Chameleon Hash* (PCH), introduced by Derler et al. [17], instantiates this pattern with Ciphertext-Policy Attribute-Based Encryption (CP-ABE). The creator attaches an access policy $\mathbb{A}$, expressed as a monotone Boolean formula over a set of attributes, and etd is encrypted under $\mathbb{A}$, so that only a party holding a satisfying attribute key can decrypt it and run Adapt. CHET plus CP-ABE is the dominant architectural paradigm in the feature-rich redactable blockchain literature; most subsequent advances (accountability, revocation, multi-authority, dynamic governance) are best understood as extensions of it.

2.3 Lattice-Based Cryptography and Post-Quantum Hardness

Lattice-based cryptography provides the most mature family of post-quantum cryptographic constructions and forms the basis for all but two post-quantum CHF schemes surveyed in this work. We recall the two central hardness assumptions.

Short Integer Solution (SIS) and Inhomogeneous SIS. Let n, m, q be integer parameters and $\beta > 0$ a norm bound. Given a uniformly random matrix $A \in \mathbb{Z}_q^{n \times m}$, the $\text{SIS}_{n,m,q,\beta}$ problem asks for a nonzero vector $z \in \mathbb{Z}^m$ with $\|z\| \leq \beta$ such that $Az = \mathbf{0} \pmod{q}$. SIS is the basis of collision resistance in lattice CHFs: finding a chameleon hash collision without the trapdoor reduces to solving SIS. Ajtai [3] showed that average-case SIS hardness follows from worst-case hardness of approximating short vector problems on n -dimensional lattices via a classical reduction that holds against adversaries of any computational model, including quantum. The inhomogeneous variant, $\text{ISIS}_{n,m,q,\beta}$, additionally fixes a target syndrome $u \in \mathbb{Z}_q^n$ and asks for a short z satisfying $Az = u \pmod{q}$.

Learning With Errors (LWE). Let $s \in \mathbb{Z}_q^n$ be a secret vector and χ a discrete error distribution over $\mathbb{Z}$. The decision-LWE $_{n,q,\chi}$ problem asks to distinguish pairs $(a_i, a_i^\top s + e_i \pmod{q})$, where $a_i \leftarrow \mathbb{Z}_q^n$ uniformly and $e_i \leftarrow \chi$, from uniformly

random pairs. LWE underpins encryption and key encapsulation in the lattice setting. Regev [41] reduced worst-case lattice problems to LWE, though via a quantum reduction, so LWE’s foundation presupposes quantum hardness of the worst-case problems rather than establishing it.

Trapdoor Sampling. The practical engine of lattice-based CHF is the ability to *sample* short preimages of lattice cosets when a trapdoor is known. Gentry, Peikert and Vaikuntanathan [21] formalised this via *Preimage Sampleable Functions* (PSFs): trapdoor functions admitting efficient, distribution-oblivious preimage sampling, drawing outputs from a discrete Gaussian distribution whose parameters depend only on a scalar width rather than the specific geometry of the secret basis. The trapdoor is a short basis of a lattice kernel; without it, finding short preimages is as hard as SIS. In a lattice CHF, the Adapt algorithm is precisely this preimage sampling operation, first instantiated concretely by Cash, Hofheinz, Kiltz and Peikert [9], whose CHF reduces collision resistance to SIS in the standard model but achieves only weak collision resistance and no key exposure freeness, a gap that later post-quantum CHF constructions (Section 4) attempt to close. Their broader contribution, a hierarchy of lattices whose trapdoors can be delegated to child lattices without loss of quality (the *bonsai tree* framework), is the standard template for hierarchical trapdoor management later adopted by identity-based schemes such as Wang et al.’s HIBCH-RS [48] and Li and Liu’s tagged CHF [31]. Micciancio and Peikert [36] substantially improved the efficiency of trapdoor generation and sampling via the *gadget matrix* construction, achieving near-optimal lattice dimension and trapdoor quality while reducing storage from quadratic to linear in the lattice dimension. The resulting machinery, called MP-trapdoors, underlies virtually all lattice-based CHF schemes surveyed in this work.

Quantum Resistance. The security of SIS and LWE rests on the hardness of approximating lattice problems such as the Shortest Vector Problem (SVP) and its variants. No polynomial-time quantum algorithm is known for these problems at the approximation factors relevant to cryptography; the best known quantum algorithms run in time exponential in the lattice dimension, improving on their classical counterparts only in the constant in the exponent rather than in the order of growth [14], which is why lattice assumptions underpin the NIST post-quantum standards [4]. This contrasts sharply with the exponential speedup Shor’s algorithm provides against discrete logarithm and integer factoring, which breaks all CHF constructions based on factoring or discrete logarithm [42].

3 Methodology

The review is designed as a structured literature survey with two objectives. First, to assemble a representative corpus of work at the intersection of chameleon hash functions (CHFs), redactable blockchain systems, and post-quantum cryptography (PQC), sufficient to characterise the current state of the field. Second, to evaluate each paper in the corpus against a common set of criteria, producing a structured comparison that makes explicit which security properties and governance features have and have not been achieved under quantum-resistant assumptions. The aim throughout is reproducibility and transparency of coverage rather than exhaustive completeness across every venue.

3.1 Search Strategy

Papers were identified through a combination of database search and iterative backward bibliography tracing. The primary databases consulted were Google Scholar, IEEE Xplore, and the IACR ePrint Archive; these were supplemented by targeted web search. Search terms were drawn from two axes:

- **Topic qualifiers:** *redactable blockchain, chameleon hash, policy-based chameleon hash, sanitizable signatures, mutable blockchain.*

- **Post-quantum qualifiers:** *lattice-based, post-quantum, PQC, quantum-resistant.*

These terms were queried both independently and in combination. The topic axis was intended to capture the full redactable blockchain and CHF-primitive literature regardless of cryptographic foundation; the PQC axis was then used to identify the post-quantum subset and to surface foundational lattice trapdoor works. Once an initial corpus was established, the reference lists of retrieved papers were traced backward iteratively until no new in-scope papers were surfaced. Forward citation search was not performed.

3.2 Inclusion and Exclusion Criteria

Papers were included in the corpus if they satisfied at least one of the following three conditions:

- (1) **Redactable blockchain scheme.** The paper proposes or evaluates a blockchain or distributed ledger construction that incorporates a chameleon hash or related equivocal commitment as its redaction mechanism, whether classical or post-quantum.
- (2) **CHF primitive.** The paper introduces or substantially extends a chameleon hash function relevant to the redactable blockchain setting, even where no blockchain system is proposed.
- (3) **Enabling post-quantum primitive.** The paper constructs a post-quantum primitive filling a named component slot in an established redactable blockchain architecture that the CHF literature does not itself supply, so that its availability bounds what post-quantum redaction schemes can be built.

Papers were excluded if they addressed blockchain redaction exclusively through non-cryptographic mechanisms (e.g. governance-only or directed-acyclic-graph-restructuring approaches with no CHF component), or if they appeared only in non-retrievable grey literature. No papers were excluded solely on the basis of publication venue, date, language, or proof methodology.

3.3 Corpus Composition

The search and selection process yielded an initial set of 42 papers. Table 2 summarises the composition by category. Of the 42 papers, 31 were accessed in full via institutional access or open-access repositories; a further 11 were identifiable through abstracts and meta-data but could not be accessed in full. These partially-accessed papers [5, 25, 28–30, 33, 38, 39, 46, 55, 58] are not included in the comparative tables (Tables 4–7).

Table 2. Composition of the identified paper set. The 31 fully accessed papers constitute the review corpus.

Category	Identified	Fully Accessed
Redactable blockchain schemes	30	22
Primitive papers	12	9
Total	42	31

3.4 Comparative Framework

Each paper in the corpus was evaluated against twenty-six criteria. Three are bibliographic (Title, Author(s), and Year, the last referring to the primary publication venue). Eight are recorded directly from each paper and need no further definition: Mathematical Structure and Hardness Assumption (the algebraic domain and the specific problem the security proof reduces to); PQ-Secure (Yes, No, Partial) and PQC Primitive Family (Lattice, Hash, Code, N/A); Redaction Granularity (Block, Transaction, Field); Blockchain Type (Permissioned, Permissionless, Both, N/A); Implementation Provided; and Platform (Hyperledger, Ethereum, Tendermint, Bitcoin, Custom, None). The remaining fifteen involve a coding judgement, and their definitions and permitted values are given in Table 3. Several are standard in the CHF and

provable security literature; others are specific to the redactable blockchain setting. The resulting comparative tables (Tables 4–7) form the empirical basis for the analysis in Section 5.

Table 3. Definitions and permitted values for the criteria in Tables 4–7 that involve a coding judgement.

Criterion	Definition	Values
Proof Model	Idealised model, if any, invoked by the security proof.	Standard, ROM, GGM, Offers Both (Standard & ROM)
Collision Resistance	Strongest CR notion formally proved, per the taxonomy of Sec. 2.1	W-CR, E-CR, S-CR, F-CR, None, No Proof Given, N/A
Indistinguishability	Redacted content indistinguishable from an original, even to a public-key holder	Yes, No
Key Exposure Freeness	Observed collisions leak no trapdoor information; recorded where a paper claims or proves the property, including under a stronger notion it names differently	Yes, No, N/A
Permission Model	Access-control model governing redaction authority; multiple values may apply	Centralized, Threshold, ABE-Policy, Identity-Based, Decentralized
Multi-Authority	Multiple independent attribute/trust authorities, rather than one	Yes, No
Accountability	Mechanism to identify who performed a redaction	No; Weak (log/signature-linkable); Strong (black-box tracing)
Revocability	A party's redaction capability can be revoked without replacing the whole trapdoor infrastructure	Yes, No
Key/Trapdoor Update	Key or trapdoor material can be reissued in place, invalidating what it replaces; constrains capability from the update onward only.	Yes, No
Dynamic Node Support	Nodes join/leave the redaction-eligible set without full re-keying	Yes, No
Forward/Backward Security	Key material held at epoch t confers no redaction capability over hashes committed at epochs $< t$ (forward), and none at epochs $> t$ (backward).	Yes, No
Time-Bound / k -time Redaction	Redactions limited by a time window or a count bound	Yes, No
Policy Privacy	Access structure governing redaction is hidden from parties who do not satisfy it. N/A where redaction is gated on key or share possession alone, with no such structure published alongside the hash.	Yes, No, N/A
Consistency Mechanism	Guarantees honest nodes converge on one canonical chain after redaction	Yes, No, N/A
Standard Model Proof	Whether a complete proof of the claimed properties is established without idealisation.	Yes, No, Partial

4 Chameleon Hash Constructions and Blockchain Deployment

Chameleon hash functions were introduced by Krawczyk and Rabin [27], who observed that trapdoor hashes, which are collision-resistant to outsiders but admit efficient collision-finding for a trapdoor holder, could serve as the basis for chameleon signatures. *Key exposure* was immediately identified as a weakness in the original construction. In the discrete-log instantiation, observing a single collision is sufficient to recover the trapdoor entirely. This is critical in the blockchain setting, where every redaction event necessarily broadcasts a valid collision to the network. Key-exposure

free (KEF) variants followed, in which the output of Adapt reveals no information about sk even after many observed collisions.

Derler, Samelin and Slamanig [16] settled the correct notion for this setting at PKC 2020 (Section 2); this survey takes F-CR as the security target for any CHF deployed in a live redactable blockchain, and evaluates the literature against it.

4.1 Foundations

Ateniese, Magri, Venturi and Andrade (2017). The redactable blockchain as a concrete system was introduced by Ateniese et al. [7], who replaced the hash linking each block to its predecessor with a chameleon hash, so that a trusted trapdoor holder can replace a block’s content while preserving the hash value seen by the rest of the chain. Redaction is at block granularity and is indistinguishable from an original block to any observer without the trapdoor. Security rests on the discrete logarithm assumption; the permission model is a single trapdoor holder, with an optional threshold variant distributing shares via secret sharing; and no accountability or policy-based access control is provided. Despite these limitations the scheme defines the architectural template that virtually every subsequent construction builds upon.

Camenisch, Derler, Krenn, Pöhls, Samelin and Slamanig (2017): CHET. The standard CHF is all-or-nothing: any holder of sk can adapt any hash ever computed under the corresponding public key, whether or not the hashing party intended to permit it. CHET [8] (Section 2.2) resolves this by requiring both sk and a per-hash ephemeral trapdoor etd , generated fresh by the hashing party, who then decides post hoc whether to release it. Beyond the standard CHF properties, CHET requires public collision resistance, private collision resistance (the long-term trapdoor holder cannot adapt without etd), and indistinguishability. *Uniqueness*, that a message has exactly one valid randomness for a given hash value, is defined as an optional strengthening rather than a requirement, a distinction that becomes decisive in Section 5.6. All constructions given are classical; the first lattice CHET is due to Klamti and Hasan [26].

4.2 The Classical Baseline

The PCH lineage. The central architectural advance over Ateniese et al. is the move from block-level to transaction-level redaction under attribute-based access control. Derler, Samelin, Slamanig and Striecks [17] introduced the Policy-Based Chameleon Hash (PCH), composing CHET with CP-ABE (instantiated with FAME [2], upgraded to indistinguishability under adaptive chosen-ciphertext attack (IND-CCA2) by a Fujisaki–Okamoto transform) so that the ephemeral trapdoor is recoverable only by a party whose attributes satisfy an access policy stored on-chain alongside the transaction. CHET plus CP-ABE is the dominant pattern in the feature-rich literature, and the classical advances that follow are all extensions of it. PCHBA [44] adds black-box accountability, tracing a modification to its author even when a coalition has pooled trapdoors into a shared decryption box. DACH [61] handles attribute authorities going permanently offline through a Delegate algorithm that hands fresh key material from a departing authority to an incoming one. TPCH [10] binds a time parameter into the hash so that redaction time is verifiable through Ver, and adds an Update algorithm re-binding a hash to a new time period, with standard model security following from composition results proved elsewhere. PRBC [52] makes the chameleon trapdoor key fully public so that policy alone gates redaction, with accountability supplied by an identity registration table. PFRB [22] is the first to hide the access policy itself, encoding it in polynomial coefficients rather than an explicit LSSS matrix. All six are classical. The mechanisms behind the three properties that matter most for the post-quantum comparison, DACH’s epoch transition, PCHBA’s accountability witness, and PFRB’s policy hiding, are examined where they are used, in Sections 5.4 and 5.6.

Threshold and decentralized governance. Four further classical schemes extend threshold governance along distinct axes. Miao et al. [35] pair a discrete-log double-trapdoor chameleon hash with a nested Strong-RSA Retrievable Homomorphic Verifiable Tag (RHVT) scheme that aggregates block tags into a constant-size global tag, cutting authentication-path storage, with a hierarchical Certificate Authority able to directly override and repair a corrupted committee member’s edits. VDCH [53] replaces the single-holder trapdoor with a Pedersen Distributed Key Generation (DKG)-style Multi-Party Computation (MPC) scheme in which the full trapdoor is never materialised and each collision share is individually verifiable, cutting redaction latency from hours to hundreds of milliseconds via a companion consensus protocol. DRBC [23] keeps the trapdoor permanently implicit in a shared bivariate polynomial, supporting dynamic committee membership through proactive secret sharing that temporarily doubles the reconstruction threshold during transitions. AWTC [34] adds accountability via a weighted Shamir Secret Sharing selection scheme and a tracer-key mechanism, roughly doubling editing throughput through non-interactive adaptation. All four are proved in the random oracle model or informally.

Alternative architectures. A handful of schemes sit outside both paradigms. Deuber et al. [18] dispense with chameleon hashes entirely, using pure proof-of-work miner voting on proposed redactions, a structurally distinct and more naturally quantum-resistant path, but one potentially misaligned with the unilateral erasure obligations (e.g. GDPR) that motivate much of this literature. Dong et al. [19] pair a standard CHF with independently-deployed Multi-Authority ABE for payload confidentiality rather than access-gated redaction, making theirs not a Policy-Based Chameleon Hash in Derler et al.’s sense. Dai et al. [13] invert the trust model, making the trapdoor fully public and bootstrapping security through an outer SHA-256 commitment instead, an instantiation substitutable for a post-quantum CHF such as Li and Liu’s [31] without architectural change.

4.3 Post-Quantum Primitives

As established in Section 2, Cash et al.’s foundational lattice CHF [9] realises collision-finding from SIS hardness but achieves only W-CR with no KEF. The following papers attempt to close this gap to F-CR and KEF, each making a different trade-off between assumption strength, proof model, and the need for heavy cryptographic tools such as Non-Interactive Zero-Knowledge proofs (NIZKs).

Derler, Krenn, Samelin and Slamanig (SCN 2020): F-CR from LPN. This paper [15] makes two contributions: it weakens the assumptions required to achieve F-CR, and in doing so produces the first post-quantum instantiation of an F-CR CHF. The PKC 2020 construction of F-CR used a perfectly correct indistinguishability under chosen-plaintext attack (IND-CPA) secure PKE scheme together with an simulation-sound extractable NIZK. This paper identifies that Public-Key Encryption (PKE) was doing more work than necessary: its binding and hiding properties can be supplied by a plain non-interactive commitment scheme, a strictly weaker primitive. Replacing PKE with commitments preserves both full collision resistance and indistinguishability while substantially relaxing the assumptions needed for a post-quantum instantiation.

The instantiation under the Learning Parity with Noise (LPN) assumption builds on a commitment scheme over binary linear codes, with zero-knowledge proofs constructed via syndrome decoding. The key pair is structured as an LPN instance, and the OR-language proof exploits the fact that both branches share the same algebraic form. This makes it a natural fit for the syndrome-decoding framework. Security is argued in the random oracle model via the Fiat-Shamir transform. The paper also introduces *randomness unforgeability*, a new property between trapdoor-commitment-level security and uniqueness, and proves it is strictly separated from both.

The limitation is proof overhead. Every hash in a deployed chain must carry a syndrome-decoding zero-knowledge proof with non-trivial per-round soundness error, and every node must verify it on every block, against a single norm check for a lattice construction. The scheme is a proof-of-concept that PQ F-CR is achievable from simple assumptions, not a deployment-ready primitive.

Wu, Ke and Du (2021): KEF from ISIS, Without NIZK. Wu et al. [50] address the absence of key-exposure freeness in all prior lattice CHF constructions. They present two distinct single-trapdoor KEF CHF schemes, both lattice-based, that avoid PKE and NIZK entirely, making them more compact and efficient than any construction in the PKE+NIZK paradigm.

Scheme 1 exploits a structural analogy between chameleon hash collision-finding and lattice signature computation. By combining the twin-signature framework of Naccache–Pointcheval–Stern with Ducas et al.’s bimodal Gaussian rejection sampling, the authors construct a scheme where computing a collision is equivalent to forging a lattice signature (hard under ISIS). Security is proved in the Generic Group Model (GGM), and notably the hash function H needs only to be collision-resistant, not a full random oracle. Scheme 1 also supports a fully distributed key generation and collision computation via a semi-honest secure multi-party sum protocol, meaning no single party ever holds the complete trapdoor. This is a governance feature not found in any other lattice CHF construction in this survey.

Scheme 2 is built with GPV lattice trapdoors, structured similarly to a Pedersen commitment, and reduces to SIS. Security is proved in the random oracle model. The primary limitation is that Scheme 1’s proof relies on the idealised GGM, Scheme 2’s proof is ROM-based, neither achieves a standard model proof, and no implementation is provided.

Wu et al.’s two lattice-based constructions also map onto two distinct redaction mechanisms for blockchain governance, making this the most governance-complete post-quantum scheme in the survey. It is the only post-quantum construction to address both threshold key management and transaction-level accountability. A threshold secret-sharing approach paired with Scheme 1’s semi-honest MPC protocol distributes key generation and collision computation so that no single party ever holds the complete trapdoor, yielding indistinguishable redaction. A consensus-based voting strategy, applicable to both schemes, provides public accountability while remaining compatible with permissioned, consortium, and permissionless architectures. Distributed key management for Scheme 2 specifically remains an open problem.

Li and Liu (2024): F-CR from SIS in the Standard Model. Li and Liu [31] provide the most practically motivated post-quantum CHF construction in the survey, explicitly framing their work as a response to the NIZK overhead of the Derler et al. SCN 2020 scheme. Their key observation is that the blockchain setting provides a structural property that generic CHF applications do not, that is, every block or transaction has a unique, one-time identifier (timestamp, block position, transaction ID) that can serve as a *tag* for the hash computation. Exploiting this, they define a *tagged CHF* in which the security notion is weakened from full F-CR to *restricted collision resistance* (r-CR), which restricts the adversary’s adaptation queries and winning condition rather than the collisions it is allowed to observe: the adversary may still see many collisions under the tag it later attacks, but each tag is bound to a single hash value and each adapted message is fresh. They prove r-CR and F-CR equivalent when tags are genuinely one-time, a mode a blockchain supplies for free.

The construction uses MP-trapdoors [36] directly. The tag is concatenated with the message and hashed under a matrix assembled by homomorphic evaluation of a pseudorandom function (PRF) circuit, and the hash is computed as $h = [A \mid A_{\text{prf}}] \cdot e$ for a short e . Collision-finding is a trapdoor delegation followed by a single preimage sampling call, with no NIZK required at any step. The security proof is in the standard model, reducing r-CR to SIS hardness and to

PRF pseudorandomness, hence to SIS and LWE under a lattice PRF. A second construction trades the standard model for a random oracle in place of the homomorphic evaluation, buying a tight reduction to SIS alone and better efficiency.

The scheme does not claim KEF, but KEF is the weaker property: it guarantees security only while no collision under the target tag has been revealed, whereas r-CR tolerates polynomially many. What the security does rest on is the one-time-tag assumption, and whether that substitution is sufficient for all blockchain deployment scenarios depends on whether one-time tags can be guaranteed in practice.

The construction is also applied directly to the block-level setting, using each block’s position identifier as the one-time tag – a natural fit, since block positions are non-repeating by construction, with redaction at block rather than transaction granularity and no additional governance features beyond centralised trust.

Clermont et al. (2025): Tag-Free F-CR and the Uniqueness Barrier. Clermont, Düzl , Janson, Porzenheim and Struck [12] close the gap between Li and Liu’s tagged construction and fully general F-CR by constructing the first tag-free F-CR lattice CHF. Their motivation is the sanitizable signature (SSS) setting, where the same key is reused across many hash evaluations with no external unique identifier, making Li and Liu’s one-time-tag assumption inapplicable. The result is directly relevant to redactable blockchains as well, since it provides a CHF that is usable in both settings without architectural constraints.

The construction builds on Li and Liu’s tagged CHF, stripping the one-time tag and patching the resulting security hole with a trapdoor commitment scheme. Without the tag, nothing binds the short vector opening a hash to that hash, so an adversary with adapt oracle access can take a collision it has been given and scale both openings by a small constant: linearity preserves the equality, but the hash value scales with them, so the result is a valid collision on a hash value never queried, which is precisely the full collision resistance winning condition. Committing to the opening at hash time removes that freedom. The commitment must be a trapdoor commitment, since Adapt has to equivocate its own earlier commitment, and the distributional equivalence of equivocation (DEE) property is what makes the equivocated opening indistinguishable from a fresh one. The scheme relies only on SIS hardness, a random oracle, and a trapdoor commitment; no NIZK is required in the CHF itself, and both F-CR and indistinguishability are achieved.

The paper also identifies a structural limitation of lattice CHFs with direct implications for accountability in the redactable blockchain setting. The authors prove that their own construction fails *uniqueness*, the property that a message has only one valid randomness for a given hash value, and argue that the failure is generic rather than specific to their scheme: because Adapt draws from a discrete Gaussian over a coset containing exponentially many admissible short vectors, it returns a different one on essentially every call, so multiple valid randomness values exist for every message–hash pair. They conjecture that achieving uniqueness requires trapdoorizing the hash by some means other than Gaussian preimage sampling. The consequences for accountability are discussed in Section 5.6.

Faced with this, the authors recover accountability by a route that does not pass through uniqueness at all, applying a transform of Bultel et al. that requires a verifiable ring signature and constructing the first lattice instantiation of one to instantiate it. The result is confined to the sanitizable signature setting, but it establishes that a uniqueness-free witness layer is constructible in the lattice setting.

Outside the lattice setting, the corpus contains one further post-quantum CHF, the hash-based scheme of Thanalakshmi et al. [43], built over a directed acyclic graph with security reducing to preimage resistance, which the authors argue survives Grover’s algorithm at an increased security parameter. It has not been adopted in the redactable blockchain literature and establishes only weak collision resistance. Its hash-based framing is also questionable, requiring a homomorphic hash function and pseudorandom generator, substantially stronger objects than a plain collision-resistant hash, with a natural instantiation that is essentially SIS-adjacent.

Klamti and Hasan (2024): The First Lattice RPCH. Klamti and Hasan [26] provide the only post-quantum construction at the policy-based layer in this survey. Their *Revocable Policy-Based Chameleon Hash* (RPCH) is built entirely from lattice hard problems (SIS, ISIS, and decision-LWE) and directly lifts the generic RPCH construction of Xu et al. [56] to the post-quantum setting. The paper produces two independently useful sub-contributions alongside the RPCH itself. They introduce a new lattice-based CHET, and the first adaptively secure lattice-based Revocable Attribute-Based Encryption (RABE) scheme.

The lattice CHET modifies Cash et al.’s construction to use a single public matrix and introduces a compact random binary seed as the trapdoor, recovered on demand via a random oracle, making the trapdoor small enough to encrypt under RABE, which is necessary in a practical sense since the ephemeral trapdoor must be transmitted as ciphertext. The RABE is built on Tsabary’s lattice ABE with a binary tree revocation structure; the standout claim is adaptive IND-CPA security, which every comparable prior lattice RABE lacked. The RPCH is then assembled using a Fujisaki–Okamoto-style transformation, with security of the combined scheme following from Xu et al.’s composition theorem.

The limitations are significant for deployment purposes: the construction supports only t -CNF (conjunctive normal form) access policies, a strict subset of monotone Boolean formulas; the proof requires the random oracle model despite the underlying hardness assumptions being standard; no blockchain integration or implementation is provided; and policy privacy, accountability, and dynamic node support are all absent. Nevertheless, Klamti and Hasan represent the current frontier of post-quantum policy-based CHF constructions, and their lattice CHET and RABE sub-contributions are building blocks that subsequent work can extend.

No lattice construction in this landscape combines a standard-model proof with KEF; the reasons are set out in Section 5.3, and the verdict, that what is missing is a completeness result rather than a live weakness, in Section 6.

4.4 The Post-Quantum Frontier

The corpus’s only implemented post-quantum scheme is described below. **Wang, Chen, Zhu, Li and Fang [48]** present HIBCH-RS (Hierarchical Identity-Based Chameleon Hash with Revocable Subkey), a lattice-based scheme deployed on Hyperledger Fabric, addressing the gap left by Wu et al. [50]: a theoretical construction relying on key sharing for trapdoor management, with no revocation or update mechanism.

The central novelty is the slave key mechanism. In standard hierarchical identity-based schemes a parent generates its child’s trapdoor directly and can therefore always impersonate the child, which is a problem where trapdoors represent edit authority. HIBCH-RS instead requires the child to generate its own slave key pair, a fresh lattice trapdoor bound to its identity via Full-Rank Difference encoding, which the parent then uses during delegation: the child trapdoor is bound to a private slave key the parent never sees, yet the parent retains the ability to invalidate it by issuing a new tag matrix. This gives trapdoor independence between levels while preserving top-down revocation, and the delegated key size does not grow with hierarchy depth, since the mechanism absorbs per-level expansion into a fixed-dimension structure. For private transactions, an LWE-encrypted identity verification pair is embedded into the hash computation, enforcing that only the transaction’s owner can edit their own private data. This self-management feature is novel in a post-quantum construction. Consistency, traceability, and accountability are handled through a three-block-type chain structure with majority voting on edit requests and a revocation list for malicious editors. Security proofs are in the random oracle model, with collision forgery resistance reducing to ISIS hardness, and indistinguishability proved statistically via closeness of discrete Gaussian distributions. Edit times range from 4.66 seconds (public transactions) to 8.29 seconds (private). The main limitations are the computational cost of trapdoor delegation at production-grade lattice parameters,

the absence of forward or backward security, k -time redaction bounds, and the restriction of multi-authority support to a fixed hierarchical tree rather than arbitrary attribute policies.

5 Comparative Analysis

This section identifies and analyses the principal gaps in the post-quantum redactable blockchain literature. Each paper was characterised across 26 dimensions in the tables below, covering mathematical structure and hardness assumptions, system and blockchain architecture, core cryptographic security properties, and advanced governance and lifecycle features.

5.1 Mathematical Foundations and Proof Models

Table 4 records the mathematical structure, hardness assumption, proof model, and standard-model proof status for all thirty-one fully evaluated papers. Reading across these four columns reveals several consistent patterns that shape the rest of the comparative analysis.

Symbols used throughout this sections tables: ✓ = Yes; × = No; – = N/A; ~ = Partial/unclear; W = Weak accountability; S = Strong/black-box accountability; NPG = No Proof Given.

The collision-resistance notions (W-CR, E-CR, S-CR, F-CR) follow the taxonomy of Derler, Samelin and Slamanig, as defined in Section 2.1; note in particular that $F-CR \Rightarrow S-CR \Rightarrow W-CR$ and $F-CR \Rightarrow E-CR \Rightarrow W-CR$, with F-CR the correct on-chain deployment target. Accountability (Weak/Strong) and Proof Model (Standard/ROM/Offers Both/GGM) are defined in Section 3.4.

Table 4. Mathematical structure, hardness assumptions, and proof model. Assumptions marked [†] are non-standard, being either not reducible from a well-studied problem or parameterised by adversarial query count.

Ref	Math. Structure	Hardness Assumption	Proof Model	PQ-Secure	PQC Family	Std. Model Proof
[17]	Integer Factoring	1-More RSA Inversion [†]	ROM	×	–	×
[7]	Discrete Log	DL	Offers Both	×	–	~
[31]	Lattice	SIS	Offers Both	✓	Lattice	✓
[48]	Lattice	ISIS	ROM	✓	Lattice	×
[50]	Lattice	ISIS	ROM / GGM	✓	Lattice	×
[61]	Integer Factoring	1-More RSA Inversion [†]	ROM	×	–	×
[35]	Discrete Log	DL	ROM	×	–	×
[16]	Discrete Log	DDH	Offers Both	×	–	~
[15]	Code Based	LPN	ROM	✓	Code	×
[12]	Lattice	SIS	ROM	✓	Lattice	×
[23]	Pairing Based	DBDH	Standard	×	–	~
[10]	Discrete Log	DL	Standard	×	–	~
[34]	Discrete Log	CDH	ROM	×	–	×
[53]	Discrete Log	CDH	ROM	×	–	×
[19]	Pairing Based	DBDH	ROM	×	–	×
[43]	Hash Based	Preimage	Standard	✓	Hash	~
[52]	Pairing Based	q-Type [†]	ROM	×	–	~
[26]	Lattice	SIS	ROM	✓	Lattice	~
[8]	Discrete Log	DL	Standard	×	–	✓
[22]	Pairing Based	DBDH	ROM	×	–	×
[51]	Discrete Log	DL	ROM	×	–	×
[57]	Integer Factoring	RSA	ROM	×	–	×
[11]	Pairing Based	DLIN	ROM	×	–	×
[13]	Discrete Log	DL	ROM	×	–	×
[45]	Pairing Based	q-Type [†]	ROM	×	–	×

Table 4 (continued)

Ref	Math. Structure	Hardness Assumption	Proof Model	PQ-Secure	PQC Family	Std. Model Proof
[54]	Pairing Based	DLIN	ROM	×	–	×
[37]	Discrete Log	DL	ROM	×	–	×
[24]	Integer Factoring	1-More RSA Inversion [†]	ROM	×	–	×
[56]	Integer Factoring	RSA	ROM	×	–	×
[49]	Lattice	Evasive-LWE [†]	Offers Both	✓	Lattice	✓
[44]	Pairing Based	DLIN	ROM	×	–	×

Classical dominance. The literature is overwhelmingly classical. Of the thirty-one fully evaluated papers, twenty-three rest on hardness assumptions broken by Shor’s algorithm, including discrete logarithm, CDH, DDH, DBDH, DLIN, RSA, and their variants. The eight post-quantum papers span three primitive families, six are lattice-based [12, 26, 31, 48–50], one code-based [15] (LPN-based), and one hash-based [43], the last relying on preimage resistance rather than any algebraic hardness assumption. Within the lattice papers, SIS and ISIS are the dominant assumptions. LWE appears mainly as a supporting primitive, as the PRF instantiation in Li and Liu’s standard-model construction [31] and as the base assumption of Waters, Wee and Wu’s MA-ABE [49], whose standard-model construction additionally requires evasive LWE. No post-quantum paper in the corpus proposes a full blockchain-integrated scheme from code-based or hash-based assumptions; both non-lattice PQ papers are CHF primitive papers with no on-chain deployment.

Assumption strength in the classical literature. Even within the classical papers, the strength of the underlying assumptions varies considerably. Several of the most feature-rich constructions rest on assumptions that are non-standard even in the classical sense. The 1-More RSA Inversion assumption, used by Derler et al.’s PCH [17], Zhang et al.’s DACH [61], and Jia et al. [24], is not reducible from standard RSA. The q -type assumptions used by Tian et al. [45] and PRBC [52] are query-parameterised and generally regarded as a weaker foundation than static assumptions. This tendency is most visible in the ABE-policy and accountability settings, where the expressive power demanded by fine-grained access control and public traceability has pushed constructions toward non-standard assumptions.

ROM dominance and the standard-model gap. The most consistent pattern in Table 4 is the near-universal reliance on the random oracle model. Of the thirty-one papers, twenty-two use the ROM throughout with no standard-model alternative offered, and one [50] uses the ROM and the generic group model across its two constructions. Only three papers provide a full standard-model proof: Li and Liu’s tagged CHF [31], which offers both a standard-model and a ROM construction, Camenisch et al.’s foundational CHET [8], which achieves its core security properties in the standard model, and Waters, Wee and Wu’s MA-ABE [49], whose evasive-LWE construction dispenses with the random oracle in its own plain-LWE construction requires. A further seven papers provide partial standard-model results covering some but not all components necessary for a full proof; Chen et al.’s TPCP [10] is the borderline case, using no random oracle but resting its theorems on compositions proved elsewhere. Two of the three are post-quantum (Li and Liu; Waters, Wee and Wu), and both are primitive papers with no on-chain deployment. The post-quantum literature has therefore not produced a single standard-model proof for a complete, deployed redactable blockchain scheme.

A small, coherent post-quantum assumption base. Unlike the classical literature’s spread into non-standard assumptions, the lattice papers in this survey draw on a narrow, closely related base: SIS and ISIS underpin collision-finding in five of the six. The single departure in assumption is Waters, Wee and Wu’s evasive LWE [49], the only non-standard assumption in the post-quantum corpus and one with limited cryptanalytic history to date, adopted to reach a fully decentralised MA-ABE without random oracles. The other divergence is one of proof model rather than

assumption: Wu, Ke and Du’s Scheme 1 [50] rests on standard ISIS but proves security in the generic group model, a stronger idealisation than the ROM, to obtain key exposure freeness without NIZK or PKE. The corpus’s central limitation is therefore not assumption fragmentation but its small size and reliance on idealised models, which leave little room for the parameter comparisons the classical literature supports.

5.2 System Architecture and Access Control

Table 5 records permission model, blockchain type, redaction granularity, multi-authority support, dynamic node support, and consistency mechanism for all thirty-one papers. Read at the level of the table’s binary categories, the post-quantum literature has touched every governance feature tracked in Table 5 at least once, but a ‘yes’ in these columns rarely matches the structural sophistication of its classical counterpart. Wang et al.’s HIBCH-RS [48] delegation mechanism (Section 4.4), for instance, remains single-layer, where the most elaborate classical schemes combine multiple supervisory layers at once (e.g. Miao et al.’s [35] committee-plus-overriding-Certificate-Authority structure). No post-quantum paper attempts this kind of layering.

Table 5. Permission model, blockchain type, redaction granularity, and authority distribution.

Ref	Permission Model	Blockchain Type	Redaction Granularity	Multi-Authority	Dynamic Node Support	Consistency Mechanism
[17]	ABE-Policy	Permissioned	Transaction	×	×	×
[7]	Centralized, Threshold	Both	Block	×	×	×
[31]	Centralized	Both	Block	×	×	×
[48]	Identity Based, Decentralized	Permissioned	Transaction	✓	✓	✓
[50]	Centralized, Threshold	Both	Transaction	✓	×	✓
[61]	ABE-Policy, Decentralized	Permissionless	Transaction	✓	✓	✓
[35]	Centralized, Threshold	Both	Block	✓	✓	✓
[16]	Centralized	Permissionless	Block	×	×	–
[15]	–	–	–	×	×	–
[12]	Centralized	–	–	×	×	–
[23]	Threshold, Identity Based, Decentralized	Permissionless	Transaction	✓	✓	✓
[10]	Centralized, ABE-Policy	Both	Transaction	×	×	✓
[34]	Threshold, Decentralized	Both	Transaction	✓	×	✓
[53]	Threshold, Decentralized	Permissioned	Transaction	✓	×	✓
[19]	ABE-Policy, Decentralized	Permissioned	Field	✓	×	×
[43]	–	–	–	×	×	–
[52]	ABE-Policy	Permissioned	Transaction	✓	×	✓
[26]	ABE-Policy	–	Transaction	×	×	–
[8]	Centralized	–	–	×	×	–
[22]	ABE-Policy, Decentralized	Permissioned	Transaction	✓	×	×
[51]	Threshold, ABE-Policy	Permissioned	Transaction	×	✓	✓
[57]	Centralized, Identity Based	Permissioned	Transaction	×	×	✓
[11]	Centralized, ABE-Policy	Permissioned	Transaction	×	×	✓
[13]	Decentralized	Both	Transaction	×	✓	✓
[45]	Decentralized, ABE-Policy, Threshold	Permissionless	Transaction	✓	✓	✓
[54]	Centralized, ABE-Policy	Permissioned	Transaction	×	×	✓
[37]	Centralized, ABE-Policy	Permissioned	Transaction	×	✓	×
[24]	Centralized	Permissioned	Field	×	×	✓
[56]	Centralized, ABE-Policy	Permissioned	Transaction	×	×	–

Table 5 (continued)						
Ref	Permission Model	Blockchain Type	Redaction Granularity	Multi-Authority	Dynamic Node Support	Consistency Mechanism
[49]	Decentralized, ABE-Policy	–	–	✓	×	–
[44]	Centralized, ABE-Policy, Identity Based	Permissioned	Transaction	×	×	×

Permission model concentration. Eleven papers carry a permissionless or permissionless-compatible tag in Table 5, spreading along a governance spectrum rather than a classical/post-quantum line. At the minimal end, Ateniese et al. [7], Derler et al.’s order analysis [16] (a foundations paper, not a deployed protocol), and Li and Liu’s tagged CHF [31] rely on plain centralized trust with no multi-authority, ABE-policy, or dynamic-node support; Li and Liu, though post-quantum, adds no governance beyond Ateniese et al. Wu, Ke and Du [50] occupy a middle ground with their threshold-MPC variant which lacks ABE-policy or dynamic-node support, but adds multi-authority key management and an explicit consistency mechanism. Three further papers sit at the decentralized end, Zhang et al.’s DACH [61] and Tian et al. [45] pairing decentralization with ABE-policy control and Huang et al. [23] with identity-based threshold control, a combination that remains exclusively classical among blockchain-integrated schemes. The last four fall between these poles, holding one half of that combination or neither: Chen et al.’s TPCCH [10] attaches ABE-policy control to centralized trust, AWTCH [34] and Dai et al.’s PRBFPT [13] decentralize with no policy layer, and Miao et al. [35] layer multi-authority and dynamic-node support onto centralized threshold trust. The post-quantum footholds, via Wu, Ke and Du and, more minimally, Li and Liu, are real but shallow by comparison.

ABE-policy dominance, and its narrow post-quantum exceptions. Fifteen papers use ABE-policy access control; two are post-quantum, Klamti and Hasan’s lattice RABE [26] and Waters, Wee and Wu’s lattice MA-ABE [49], but both are unintegrated primitives with no blockchain deployment (hence no applicable Blockchain Type entry in Table 5). Neither matches the expressiveness of the pairing-based schemes the classical PCH lineage assumes: Waters, Wee and Wu reach subset and DNF policies without random oracles but only under static security, and Klamti and Hasan’s RABE reaches t -CNF policies adaptively but only in the ROM, with no multi-authority capability. Neither policy class contains the other. Carrying either into a blockchain-integrated scheme remains outstanding future work, not an impossibility.

Multi-authority operation: three post-quantum paths, none ABE-based in deployment. Twelve papers achieve multi-authority operation; three are post-quantum, each via a distinct route. Wang et al. [48] use the fixed-hierarchy delegation mechanism introduced in Section 4.4; Wu, Ke and Du [50] reach it through the semi-honest MPC sum protocol (Sections 4.3), unrelated to ABE; Waters, Wee and Wu’s [49] lattice MA-ABE remains an unintegrated primitive. The property has therefore been reached three times, but never in a blockchain-integrated scheme via the ABE-based route the classical literature relies on almost universally.

Dynamic node support and the limits of the lattice toolkit. Eight papers support dynamic node membership; exactly one, Wang et al.’s HIBCH-RS [48], is post-quantum. The remaining seven are classical. The property is most fully achieved via Zhang et al.’s Delegate algorithm in DACH [61]. HIBCH-RS [48] sidesteps rather than matches this, its slave-key delegation is a parent-child mechanism within a fixed hierarchy, not the peer-to-peer redistribution of a flat authority set that DACH allows. This is an absence discussed further in Section 5.6.

Consistency mechanisms: voting versus algebraic aggregation. Consistency mechanisms fall into three types; the post-quantum literature occupies only one. Wang et al. [48] and Wu, Ke and Du [50], the two blockchain-integrated post-quantum schemes, rely on protocol-layer voting, independent of the hash’s algebraic structure. The classical literature

additionally exploits that structure directly, via multiplicative tag aggregation (Miao et al. [35]) and key/hash rotation (Delegate [61]; Update [10]). Rotation depends on exponent-level homomorphism, for which no lattice analogue is known (Section 5.6); no post-quantum scheme attempts either, voting instead.

5.3 Security Properties: What Has and Has Not Been Achieved

Table 6 records collision resistance, indistinguishability, key exposure freeness, accountability, revocability, key/trapdoor update, and forward/backward security for all thirty-one papers. Unlike Tables 4 and 5, this table shows that properties fall into roughly three tiers, from near-universal to almost entirely classical-only.

Table 6. Cryptographic security properties.

Ref	Collision Resistance	Indistinguishability	Key Exposure Freeness	Accountability	Revocability	Key / Trapdoor Update	Forward / Backward Security
[17]	S-CR	✓	✓	W	×	×	×
[7]	E-CR	×	✓	×	×	×	×
[31]	F-CR	✓	×	×	×	×	×
[48]	E-CR	✓	✓	W	✓	✓	×
[50]	E-CR	✓	✓	W	×	×	×
[61]	S-CR	✓	✓	×	×	✓	✓
[35]	W-CR	×	✓	W	✓	✓	×
[16]	F-CR	✓	✓	×	×	×	×
[15]	F-CR	✓	×	×	×	×	×
[12]	F-CR	✓	×	–	–	–	–
[23]	NPG	×	✓	W	✓	✓	×
[10]	S-CR	✓	✓	×	✓	×	×
[34]	S-CR	✓	✓	S	×	×	×
[53]	W-CR	×	✓	W	×	×	×
[19]	–	×	×	W	×	×	×
[43]	W-CR	✓	✓	×	×	×	×
[52]	NPG	✓	–	W	×	×	×
[26]	NPG	✓	✓	×	✓	✓	×
[8]	S-CR	✓	✓	S	×	×	×
[22]	S-CR	×	×	×	×	×	×
[51]	W-CR	✓	×	W	✓	×	×
[57]	–	✓	✓	W	×	×	×
[11]	S-CR	✓	×	W	×	×	×
[13]	W-CR	×	–	S	×	×	×
[45]	–	✓	×	S	×	×	×
[54]	S-CR	✓	✓	S	×	×	×
[37]	S-CR	✓	×	W	✓	✓	×
[24]	S-CR	×	×	W	✓	✓	×
[56]	S-CR	✓	×	×	✓	✓	×
[49]	–	–	–	–	×	×	×
[44]	F-CR	✓	×	S	×	×	×

Collision resistance: Of the thirty-one papers, twenty-seven report a collision-resistance result; the remaining four [19, 45, 49, 57] are marked – since their contribution is a non-CHF component evaluated independently of a chameleon hash. Because Derler, Samelin and Slamanig’s W-CR/E-CR/S-CR/F-CR taxonomy [16] postdates much of the

corpus, each paper’s security game was read against the taxonomy directly and assigned the closest notion; the CR column is therefore an analytical classification, not a transcription of authors’ own terms.

Of the twenty-seven, eleven achieve only S-CR, five only W-CR, three no formal proof, and three E-CR. Only five achieve F-CR: Li and Liu [31], Derler, Samelin and Slamanig [16], Derler et al.’s SCN construction [15], Clermont et al. [12], and Tian et al.’s PCHBA [44]. F-CR is the correct on-chain deployment target (Section 4), since redaction is public and adaptive oracle access is the realistic threat model; against it, the literature’s “near-universal” collision resistance sits mostly below the necessary bar.

Three of the five F-CR results are post-quantum, but not on equal terms. Derler et al. achieve it from LPN via a syndrome-decoding NIZK with non-trivial per-round soundness error, in the ROM; Clermont et al. achieve it tag-free and NIZK-free, but likewise in the ROM. Li and Liu’s tagged CHF alone achieves it (as r-CR, justified by blockchain identifiers’ one-time-tag structure) in the standard model, from plain SIS (+LWE via PRF), with no NIZK. This is therefore the only post-quantum result meeting the field’s target without an idealised model or heavyweight zero knowledge machinery.

Indistinguishability: a definitional target proved with varying rigour. Twenty-two of thirty-one papers achieve indistinguishability, eight do not, and one (Waters, Wee and Wu [49]) is marked –. Waters, Wee and Wu prove ordinary ciphertext indistinguishability for their MA-ABE, a different game from the CHF/CHET notion tracked here. The eight negative cases simply omit the property from their published analysis, with one exception. In Dai et al. [13], edited blocks are marked by design for auditability, a deliberate trade-off rather than a gap.

Among post-quantum papers, seven achieve it. Indistinguishability is inherited automatically by any CHET-derived construction, since it forms part of the primitive’s own definition [8]; single-trapdoor constructions carry no such guarantee and must prove it independently, which they do noticeably less often in this predominantly classical setting. Six of the seven PQ positives are single-trapdoor constructions proving it via a construction-specific reduction; Klamti and Hasan is the exception, since their RPOCH’s new lattice CHET lets the proof follow the CHET template directly.

Key exposure freeness: Sixteen of thirty-one papers report key exposure freeness, twelve do not, three are –. The high count reflects a structural feature of the literature, not a surfeit of rigorous proofs. KEF, as Ateniese and de Medeiros [6] first defined it in response to Krawczyk and Rabin’s [27] original discrete-log construction, conflates two requirements that coincide there almost by definition, trapdoor indistinguishability (a collision reveals nothing about sk) and collision non-transferability (a collision enables no further unauthorised collisions). These are decoupled in the lattice setting: sampling a short preimage can leak nothing about the trapdoor basis while still admitting algebraic shortcuts to new collisions, as in Clermont et al.’s rescaling attack [12]. Both subproblems are nonetheless absorbed elsewhere rather than proved standalone: F-CR already implies collision non-transferability, and adapt-oracle-based collision resistance covers the trapdoor-indistinguishability half as well, Camenisch et al. noting that their own notion is strictly stronger than key exposure freeness as originally defined [8]. Where CHET/PCH-lineage papers report the property they therefore do so as an inherited consequence rather than via a dedicated theorem, and where they omit it the omission is terminological; the lineage divides roughly evenly on the label, which is why the column records assertion rather than architecture. Among post-quantum papers, only Wu et al. [50] formally proves KEF, and only under idealised models (GGM, ROM); no post-quantum paper achieves a standard-model KEF proof, and neither Li and Liu [31] nor Clermont et al. [12] (the two constructions that would most naturally need one) claim it.

Accountability: absent in the post-quantum literature. Six papers achieve strong accountability, thirteen weak, ten none, two –. PCHBA [44] remains the strong paradigm case, with black-box recoverable tracing needing no cooperation from the accused, later extended to the permissionless setting by Tian et al.’s Dynamic Proactive Secret

Sharing (DPSS)-based committee [45]. AWTCH’s tracer-key approach (Section 4.2) is formally strong but depends on the multiplicative structure of the underlying prime-order group, with exponents in $\mathbb{Z}_p$, which has no lattice analogue (Section 5.6). The remaining strong cases reach it by narrower means: attribute-based traitor tracing bound to an existentially unforgeable identity-based signature (Xu et al. [54]), a fully public trapdoor under which every edit is marked and attributable by construction (PRBFPT [13]), and SSS-context Proof/Judge algorithms rather than a blockchain deployment (Camenisch et al. [8]). Among the weak cases, VDCH [53] and Wang et al.’s HIBCH-RS [48] trace identity through protocol participation or an on-chain log rather than cryptographically, as do the two schemes whose tracing rests on an authority-held registry: PIRB’s [57] consortium-managed mapping from virtual to real identities, and PRBC’s [52] identity registration table. Every strongly accountable scheme is classical; the two post-quantum papers reaching weak accountability, Wang et al. and Wu et al. [50], do so through protocol-level mechanisms rather than any cryptographic binding of redaction to author.

Revocability and key/trapdoor update: classically mature, and typically bundled together. Nine papers achieve revocability and eight achieve key/trapdoor update, with near-total overlap; seven achieve both. Three diverge: DACH [61] has key update without revocability, while TPCCH [10] and RBFAC [51] have revocability without update.

The bundling is unsurprising, three unrelated mechanisms (binary-tree RABE broadcast [26, 56], per-attribute blinding refresh [37], stateful Revoke [24]) all conflate the two properties for the same reason. Once a scheme pushes fresh key material to legitimate holders, withholding it from one party costs nothing extra. DACH never builds an exclusion mechanism at all; TPCCH and RBFAC are the converse, adding revocability without update.

Among post-quantum schemes, only Wang et al.’s HIBCH-RS [48] achieves either property in a blockchain-integrated scheme (both, via a single slave-key call, Section 4.4); Klamti and Hasan carry both at the primitive level only. The post-quantum showing is therefore a single data point rather than a trend.

Forward and backward security: the least-achieved property in the survey, and entirely classical. Forward and backward security is achieved by exactly one paper, DACH [61], and no post-quantum construction. DACH’s guarantee belongs to its RABE ciphertext layer, not the chameleon hash: Delegate rotates attribute-authority key material across epochs, but the long-term CHF trapdoor itself is never rotated, so at the trapdoor layer capability remains unbounded in both directions. TPCCH’s Update falls short of the property, rotating no key material at all and instead invalidating hashes not refreshed by a deadline. That DACH operates one layer up is also why the property should not be conflated with revocability: revocability expires one party’s credential within a fixed infrastructure, while forward/backward security requires the infrastructure itself to rotate with old and new material decoupled.

The post-quantum absence is total here. Section 5.6 locates the difficulty in the trapdoor layer, and identifies the access-control layer, where the classical scheme operates, as the more tractable target.

5.4 Advanced Governance Features and Implementation

Table 7. Advanced redaction features and implementation details.

Ref	Time Bound / k -time Redaction	Policy Privacy	Impl. Provided	Platform	Paper Scope
[17]	×	×	✓	Custom	BC application
[7]	×	–	✓	Bitcoin	BC application; foundational

Table 7 (continued)

Ref	Time Bound / k -time Redaction	Policy Privacy	Impl. Provided	Platform	Paper Scope
[31]	×	–	×	None	CHF primitive (PQC: lattice)
[48]	×	–	✓	Hyperledger	BC application
[50]	×	–	×	None	BC application
[61]	×	×	✓	Bitcoin	BC application
[35]	×	–	✓	Custom	BC application
[16]	×	–	×	None	CHF primitive & analysis
[15]	×	–	×	None	CHF primitive (PQC: code)
[12]	–	–	×	None	CHF primitive (PQC: lattice)
[23]	×	–	✓	Custom	BC application
[10]	✓	×	✓	Custom	BC application
[34]	×	–	✓	Custom	BC application
[53]	×	–	✓	Tendermint	BC application
[19]	×	×	✓	Tendermint	BC application
[43]	×	–	×	None	CHF primitive (PQC: hash)
[52]	×	×	✓	Tendermint	BC application
[26]	×	×	×	None	CHF primitive (PQC: lattice)
[8]	×	–	×	None	CHF primitive; CHET foundational
[22]	×	✓	✓	Custom	BC application
[51]	✓	✓	✓	Custom	BC application
[57]	×	✓	✓	Custom	BC application
[11]	✓	×	✓	Custom	BC application
[13]	×	–	✓	Ethereum	BC application
[45]	×	×	✓	Custom	BC application
[54]	×	×	✓	Custom	BC application
[37]	×	×	✓	Custom	BC application
[24]	×	–	✓	Bitcoin	BC application
[56]	×	×	✓	Custom	CHF primitive
[49]	–	×	–	None	MA-ABE primitive (PQC: lattice)
[44]	×	×	✓	Custom	BC application

Time-bound and k -time redaction: three mechanisms, no shared design. Three of the thirty-one papers constrain redaction by time or count, TPCB [10], RBFAC [51], and CDEdit [11]. All are classical. each gates redaction via a structurally distinct bound rather than one recurring idea. TPCB’s Update binds validity to a time period that must be actively refreshed; CDEdit bounds by count via a four-tier, downgradable token hierarchy; RBFAC combines both axes through a Token Committee issuing capped, time-limited tokens. Miao et al.’s fixed-size cycles [35] are a storage and proof-cost parameter rather than a bound on redaction. No post-quantum construction attempts any of the three.

Policy privacy: real classical progress, and a total post-quantum absence. Policy privacy is achieved by only three of the thirty-one papers: Guo et al.’s PFRB [22], Xu and Li’s PIRB [57], and Wu et al.’s RBFAC [51]. Thirteen papers record no policy privacy, and a further fifteen are marked – rather than ×, since the property is only meaningful for schemes that publish, alongside each hash, a creator-specified object determining who may redact it. Schemes gating redaction on key or share possession alone, whether centralised, threshold, or identity-based, have no such object and hence no policy to hide. All three positive results are entirely classical.

The property is strictly stronger than standard ciphertext-policy ABE security, which hides the plaintext under a policy f but not f itself; in every deployed PCH-family scheme here, that policy sits on-chain permanently as plaintext metadata. This is a privacy failure (e.g. in IoT or healthcare deployments encoding health conditions or roles) independent of whether redaction ever occurs.

The three reach policy hiding through distinct mechanisms. PFRB [22] hides a Boolean access structure as the coefficients of a Newton-interpolation polynomial (rather than an explicit Linear Secret-Sharing Scheme (LSSS) matrix), with the zeroth coefficient hiding the trapdoor-protecting key, scaling linearly rather than exponentially with policy size. PIRB [57] is the exception to the ABE pattern, its policy being a flat per-transaction authorised-editor set rather than an attribute formula, hidden as the roots of a polynomial published only as blinded coefficients and padded to a fixed public size, so that neither membership nor policy size leaks and only an authorised identity's own evaluation yields a decryption key; it is also the only one of the three combining policy privacy with cryptographic accountability, via virtual identities letting consortium nodes deanonymise a malicious editor. RBFAC [51] closes a gap the other two share, that is, a party who can decrypt the trapdoor ciphertext can silently re-encrypt it under a different policy. By treating editing and policy modification as distinct, separately credentialed operations, bundled with revocability and k -time controllability, this paper is the most feature-complete of the three.

All three are informal or partial in their policy-hiding arguments, with no full reduction to a named hardness assumption, and none is post-quantum; no PQ construction, implemented or primitive, attempts policy privacy in any form. Together with the time-bound/ k -time findings above, the two rarest properties in Table 7 are also the two most recent (both classical work from 2022 or later) suggesting the field's second wave of governance refinement has not yet had a post-quantum counterpart.

Implementation: a post-quantum implementation gap layered on top of a formal one. Twenty-two of the thirty-one papers provide an implementation, eight do not, and one is –. Among the twenty-two, custom testbeds are the majority; the rest cluster on four platforms, three on Bitcoin, three on Tendermint, one on Hyperledger Fabric, and one on Ethereum.

The post-quantum split is the striking result underneath these totals: seven of the nine unimplemented (– and $\times$) papers are post-quantum [12, 15, 26, 31, 43, 49, 50]; the two classical exceptions, Derler et al. [16] and Camenisch et al.'s CHET [8], are both primitive-level contributions rather than deployed systems, the former pairing its relations analysis with a generic F-CR construction costed analytically rather than benchmarked. Wang et al.'s HIBCH-RS [48] is therefore the only post-quantum scheme with any deployment, and the unimplemented post-quantum papers largely coincide with those lacking a standard-model proof (Section 5.1); the two full standard-model results [31, 49] are the exceptions, well-proved but still unimplemented. The post-quantum corpus therefore has no single paper both rigorously proved and empirically validated.

5.5 The Post-Quantum Feature Frontier: A Summary

Read horizontally rather than feature by feature, Tables 4 through 7 yield a compact quantitative picture of the post-quantum literature. Six of the eight post-quantum papers are unintegrated primitives: constructions of a CHF, CHET, or ABE scheme with no accompanying redaction protocol, consensus interaction, or on-chain verification machinery [12, 15, 26, 31, 43, 49]. Li and Liu [31] come closest to an exception, applying their tagged CHF directly to block-level redaction using block position as the one-time tag, but supply no protocol or governance layer above the primitive. Only two propose blockchain-integrated schemes, these being Wang et al.'s HIBCH-RS [48] and Wu, Ke and Du's threshold construction [50]. Exactly one post-quantum paper is implemented (Wang et al. [48], on Hyperledger

Fabric), and two carry a full standard-model proof (Li and Liu [31]; Waters, Wee and Wu [49]), with no overlap: Wang et al. is proved only in the random oracle model, and neither standard-model paper reports an implementation. The post-quantum corpus therefore contains no construction that is simultaneously rigorously proved and empirically validated, a conjunction the classical literature comes closer to achieving.

Post-quantum standing by property. Table 8 sorts the feature columns of Tables 4–7 into three tiers: parity or near-parity with the classical literature, partial footholds reached only once or twice and via thinner mechanisms, and total absence.

Table 8. Post-quantum standing by property, relative to the classical literature.

Property	PQ Status	Note
Collision resistance (F-CR)	Parity+	3/5 F-CR results are PQ [12, 15, 31]; Li-Liu is the only standard-model F-CR result of any kind.
Indistinguishability	Near parity	7/8 PQ papers achieve it [12, 15, 26, 31, 43, 48, 50], comparable to the classical proportion.
Multi-authority	Partial	3 PQ papers [48–50], never via the ABE route classical schemes rely on; only 2 blockchain-integrated.
Dynamic node support	Partial (minimal)	Achieved once [48], via parent-child delegation in a fixed hierarchy, not DACH-style flat-set redistribution.
Revocability / key update	Partial (minimal)	2 PQ papers; only Wang et al. [48] blockchain-integrated, Klamti-Hasan [26] at the primitive level only.
Weak accountability	Partial (minimal)	Achieved twice [48, 50], via protocol-level audit/voting rather than cryptographic binding.
Consistency mechanism	Partial	Both deployed PQ schemes [48, 50] rely on protocol-layer voting; no algebraic aggregation as classically.
Strong accountability	Absent	No PQ construction, primitive or deployed.
Forward/backward security	Absent	No PQ construction.
Policy privacy	Absent	No PQ construction.
Time-bound / k -time redaction	Absent	No PQ construction.

A “yes” or “partial” entry in this table does not imply comparable structural depth. Where the classical literature redistributes a flat authority set across epochs [61] or aggregates consistency proofs algebraically [35], the post-quantum literature offers fixed-hierarchy delegation [48] or protocol-layer voting [50]. The absent rows, together with the partial footholds that rest on the same classical machinery, are taken up in Section 5.6.

5.6 Two Structural Obstructions

The absent properties do not share a single explanation. Some are *incidental*: missing because the work has not been attempted, not because any structural feature of the lattice setting is known to forbid it. Others are *structural*: the classical literature’s successes rest on identifiable algebraic machinery, and that machinery depends on properties of pairing-friendly or discrete-log groups for which no lattice analogue is known, so closing them requires new constructions rather than adaptation of existing ones. Two pieces of classical machinery account for every property in the second class, the epoch-transition machinery behind forward and backward security and the witness layer behind strong accountability, and each is taken in turn below. Neither is a proved impossibility for the general class: one is a proved limitation of the only lattice CHF of its kind together with an argument for its generality, and the other is the absence of any known technique.

Epoch transition. The classical exemplar is Zhang et al.’s Delegate algorithm in DACH [61]: when an attribute authority departs, Delegate updates every affected on-chain ciphertext by multiplying in a correction factor of the form $g_2^{s'(\beta'_i - \beta_i)}$, computed directly from the old and new key material without decrypting anything. This works because exponent

arithmetic in a bilinear group is ordinary $\mathbb{Z}_p$ arithmetic: the relationship between old and new keys is a scalar difference, and that difference can be applied to existing commitments as an exact algebraic identity. The same exponent-level arithmetic underlies TPCCH's re-binding of hash tuples to new time periods [10]; Miao et al.'s tag aggregation [35] folds a redacted block's tag into the global tag by group multiplication alone, in a hidden-order RSA group. Because on-chain hash values are immutable, each block committing to its transactions' Merkle root and its predecessor's hash, any epoch mechanism must relate new key material to old commitments rather than recompute history. This splits the problem in two: *Subproblem A* (the CHF trapdoor layer) asks whether trapdoor material can be rotated so that each epoch's material adapts only the hashes committed under it, old material losing capability over what follows and new material gaining none over what precedes it; *Subproblem B* (the access-control layer) asks the same question of the trapdoor ciphertexts. DACH solves only Subproblem B; even classically, the long-term CHF trapdoor is never rotated.

Consider now the natural lattice translation of this machinery. In a lattice CHF, adaptation means finding a *short* vector $\mathbf{r}'$ satisfying a syndrome equation $\mathbf{A}^{(t)} \mathbf{r}' = \mathbf{u} \pmod{q}$ with $\|\mathbf{r}'\| \leq \beta$, and the analogue of "relate old and new keys by a computable correction" would be a map carrying solutions under one epoch's matrix to solutions under another. The syndrome equation is itself linear, so solutions do transport; what does not transport is the norm bound, and this bound is essential, since SIS-hardness of finding short solutions without the trapdoor is precisely what supplies collision resistance. A correction preserving shortness in both directions hands equivalent sampling capability to both epochs, defeating the purpose of the transition; one that does not preserve it pushes transported vectors past the verification bound. No lattice mechanism reconciling the two is known, and the classical machinery offers no guidance, having never faced the constraint.

This is why a single classical mechanism is missing from three table columns at once: forward/backward security, Delegate-style dynamic membership over a flat authority set, and rotation-based consistency, which the blockchain-integrated post-quantum schemes replace with protocol-layer voting [48, 50], pushing consistency up into the consensus layer rather than solving it cryptographically. The difficulty is nonetheless confined to Subproblem A. In Subproblem B the correction is applied to the ciphertext itself, constrained by a decryption-noise bound rather than SIS-shortness. Subproblem B is accordingly the more tractable half, and the half DACH actually solves: the obstruction attaches to the full two-layer property, not to the single-layer version the classical literature has achieved.

The witness layer. The classical witness layer depends silently on *uniqueness*: for a given message and hash value there exists exactly one valid randomness, so the party who produced a particular collision output is uniquely identified as the party who used the trapdoor. All three classical accountability constructions separate the collision computation from a cryptographic record of who performed it, and it is this record, resting on uniqueness, that carries the accountability proof. PCHBA's homomorphic Schnorr signature [44] threads through the shared ephemeral trapdoor, publicly linking original and modified transactions to the same trapdoor holder (Section 4.2). AWTCH [34] encrypts an aggregated threshold Schnorr signature under a tracer key, recovering the modifier set because the aggregate verification key factors as the product $\prod_{i \in I} pk_i$ in $\mathbb{G}$, with the corresponding secrets summing in $\mathbb{Z}_p$. VDCH's per-share Schnorr NIZK [53] provides the same guarantee at the level of individual collision shares. In each case, the argument that the recorded party and no other is responsible for a given modification presupposes that the collision output could only have been produced one way. This is the property the lattice setting does not appear able to provide:

Proposition 5.1 (Uniqueness Barrier under Preimage Sampling). *Let CH be a lattice chameleon hash whose Adapt algorithm returns a discrete Gaussian preimage sampled over a coset of the lattice, in the manner of Gentry, Peikert and Vaikuntanathan [21]. Then CH does not satisfy uniqueness. The coset contains exponentially many vectors within the norm*

bound and the sampler returns a distinct one on each call with overwhelming probability, so a party holding the trapdoor can produce a second valid randomness for any message–hash pair it has already opened; correctness guarantees both are accepted by Ver. Clermont et al. [12] prove this for their own construction and argue that the obstruction is generic to the paradigm rather than an artefact of any particular scheme; every lattice CHF in this survey falls within the class.

Any accountability argument that links a modifier to a specific collision randomness, including direct adaptations of the PCHBA [44], AWTCH [34], and VDCH [53] strategies, is invalid in any lattice CHF of this class: an adversary who produced one valid randomness can produce another and repudiate the first.

Two further barriers close the remaining adaptation routes. PCHBA’s black-box tracing requires a traceable ABE, which exploits the asymmetry of Type-III pairing groups to embed recoverable identities in user keys; no lattice ABE with an analogous mechanism is known, including the Waters–Wee–Wu construction [49]. And both PCHBA’s witness thread and AWTCH’s tracer depend on Schnorr homomorphism, the relation $vk = g^{sk}$ under which two verification keys in $\mathbb{G}$ differ by a factor whose exponent is a known scalar in $\mathbb{Z}_p$, and which is therefore publicly checkable. Here the lattice analogue is not absent but approximate: a lattice verification key is an ISIS syndrome Az for short z , and such keys compose additively, since $Az_1 + Az_2 = A(z_1 + z_2)$. What does not compose is shortness: the norm grows with each operation, and the tracing constructions require unboundedly many. The accountability barrier therefore terminates in the same constraint as the epoch-transition barrier above, which is the nearest thing to a single explanation the structural class admits.

6 Open Problems and Research Directions

The gaps identified above are not equally hard, and the distinction matters more for setting a research agenda than the gaps themselves do. Two of the six items below are structural in the sense of Section 5.6, three are incidental, and one, the narrowest, is a completeness gap rather than an absence. For the incidental items the absence of a known obstruction is not proof that none exists, the claim being the weaker one that no barrier analogous to the structural ones is currently visible and that a credible adaptation path from the classical mechanism can be sketched. They are ordered by dependency: the first is a prerequisite for the two that follow it, and the remainder are independent both of that chain and of each other.

- (1) **ABE-policy deployment (incidental).** An enabling primitive now exists, at reduced policy expressiveness (Section 5.2); what is outstanding is the integration work the classical PCH literature built around FAME [2] and its successors, the redaction protocol, trapdoor encryption, and on-chain verification machinery. Waters, Wee and Wu’s MA-ABE [49] is the natural base construction, restricting on-chain policies to subset and DNF form, with Klamti and Hasan’s lattice CHET [26] supplying the primitive whose ephemeral trapdoor would be encrypted under it, in place of the t -CNF RABE they pair it with. This is a prerequisite for the two items that follow, since there is no policy to hide and no access-control layer to rotate until a lattice policy-based redaction scheme exists.
- (2) **Policy privacy (incidental, with less certainty).** The three classical schemes that achieve it, PFRB [22], PIRB [57], and RBFAC [51] (Section 5.4), encode the access structure in polynomial coefficients rather than an explicit LSSS matrix, a strategy with no obvious dependence on pairings and well understood as secret sharing over polynomial evaluations in the lattice setting. The open question is whether the blinded-evaluation step, where the discrete-log and pairing structure actually enters, survives translation. PFRB’s linear-in-policy-size Hash is the efficiency baseline any successor construction should be measured against.

- (3) **Forward/backward security (structural).** An epoch-transition mechanism for the RABE layer alone is a well-posed, plausibly achievable target, since Subproblem B is constrained by a decryption-noise bound rather than SIS-shortness (Section 5.6). This would match rather than fall short of the classical state of the art, since DACH [61], the one classical scheme that achieves this property, also solves only Subproblem B.
- (4) **Accountability (structural).** Accountability is entirely a property of the witness layer that records who performed an adaptation, separate from the collision layer itself, and its dependence on uniqueness (Proposition 5.1) is contingent rather than necessary: classical schemes rely on it only because the discrete-log setting supplies it for free. The target is therefore a witness layer that binds a modifier to an adaptation event without binding them to any particular randomness value. Clermont et al.’s verifiable ring signature [12], the first lattice instantiation of one, is the natural starting point, having been used for exactly this purpose in the sanitizable signature setting (Section 4.3).
- (5) **Standard-model key exposure freeness (a completeness gap).** This is the narrowest gap in the survey: F-CR and the adapt-oracle collision-resistance notions of the CHET lineage together discharge the two concerns KEF was designed to address for the great majority of deployments (Section 5.3), so what is missing is a completeness result rather than a live vulnerability. It matters chiefly for a single-trapdoor lattice CHF used over a long horizon, where a conservative implementer would want a guarantee, independent of any idealised model, that accumulated collision outputs give no traction against the trapdoor basis. Li and Liu’s tagged CHF [31] is the natural base construction, being the only single-trapdoor lattice CHF in the corpus with a standard-model collision-resistance result to build on.
- (6) **Time-bound/ k -time redaction, and empirical validation (incidental).** Neither requires new cryptography. Two of the three classical time-bound mechanisms [11, 51] are protocol-layer credentialing schemes that should compose with a lattice CHF and lattice signatures unchanged, and benchmarking the existing proved constructions is engineering effort rather than construction. They are placed last because neither depends on anything else in this list, not because they are the least consequential: the absence of any post-quantum construction both rigorously proved and empirically validated is among the corpus’s more serious shortcomings.

7 Conclusion

A single observation runs through the structural gaps. The classical literature solved these properties using algebraic features, bilinearity, Schnorr homomorphism, and uniqueness, that are unavailable in the lattice setting, so progress depends on re-deriving the properties from lattice-native primitives rather than porting the mechanisms that deliver them classically. The remaining items need no such re-derivation, and are outstanding rather than obstructed; the corpus’s most serious shortcoming is meanwhile not a missing property at all, but the absence of any construction both rigorously proved and empirically validated.

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